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Programmering på Cybersikkerhed 1. Semester – Mandatory 2

Introduction

Good job solving your previous assignment. We have started implementing your solutions into our pipeline.

Unfortunately, "Øresund Industrial A/S" has been heavily targeted recently, causing even more alerts, and the program you have built for us is no longer sufficient for our needs.

We now require your help again to sort through the incidents, and rather than manually sending you the incident JSON file, we will give you live access to our incident management API.

You will be required to make a program that can handle live connections, as well as store data in a database.

Our API is in beta and might not be stable, so you are required to make error handling and input validation as part of your assignment.

Description

We will provide a URL to our API endpoint. You will have to request a token. You will then have to request the incident data directly from the API.

The API returns the JSON format that you are already familiar with, but we are no longer working with CSV files as output. You will be required to make an appropriate database structure in an SQL database. We will accept SQLite files as the output of the program.

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Program requirements

- Your program must parse the JSON incident data so that you can work with it as a JSON object in Python
- We expect you to implement error handling where appropriate, and especially on the request and data storing part, to avoid database corruption.
- As we don't keep track of sessions, you must manage your token locally. Either by storing it or requesting every time you run the program.
- We expect your program to be readable and follow the PEP8 convention as much as possible.
    - You are allowed to use generative AI to help you document the program – just be aware that you are responsible for the final output of the documentation. Make sure to read it thoroughly and check for accuracy.
    - If you decide to use generative AI as part of your documentation, for example as a markdown file, please label the file as generated by AI.
- It is up to you to design the database structure, but we recommend following a similar structure with column names similar to the output of the CSV files from your previous assignment. These are columns we would expect to see at minimum:
    - alertID, incidentId, category, machineID, firstSeen, timestamp, detectionSource

Modules

You will most likely be using the "json" module as well as the "requests" module. You will also need an appropriate database module. For SQLlite we recommend the "sqlite3" module. If you want to connect to a live database or a docker container, you might want to use something like the "mysql.connector" module.

sqlite3 documentation - https://docs.python.org/3/library/sqlite3.html

mysql.connector documentation - https://dev.mysql.com/doc/connector-python/en/connector-python-example-connecting.html

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Requirements for deliverable

The project should be completed in groups.

Handing in the project is a requirement to attend the exam.

Handing in will be done through **itsLearning,** where you will also be able to register your groups. The deliverable will be a zip file containing the following:

1. Source code for the program following the specifications as described above
2. A presentation of about **10 minutes** to be presented as part of the project, the presentation could be about one or several of the following topics:
    a. Showing the source code and explaining the choices you made
    b. A short demo of your program in action
    c. Challenges you had to solve as part of your project
    d. Ideas on how to improve your program in the future